How the Zodiac Z340 Cipher Was Finally Broken

SPEAKER NOTES · 50 MIN · NUMBERS MATCH THE SLIDE FOOTERS · DROPPABLE = SKIP IF BEHIND

1 Title slide

Abstract: four cryptograms, Z408 fell in a week (Hardens, 1969), Z340 held out 51 years until Oranchak, Blake and Van Eycke, December 2020. Today: how do you know an answer is unique, how do you search for it, and why Z340 was so much harder.

PROLOGUE

The story

at 0:00

3 The Zodiac

Keep this brief and sober. Lake Herman Road (Dec 1968), Blue Rock Springs (Jul 1969), Lake Berryessa (Sep 1969), Paul Stine in Presidio Heights (Oct 1969). The letter shown is from June 1970 and contains Z32 at the bottom.

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4 Z408: solved in a week, by a schoolteacher

Z408 was mailed in three parts to the Vallejo Times-Herald, SF Examiner and SF Chronicle, with the demand that it be printed on the front page: 'In this cipher is my idenity.' 408 symbols, 54 distinct ones. The final 18 letters EBEORIETEMETHHPITI have never been convincingly read. The misspellings are in the original. Frequency analysis plus a crib (a guessed word) is the 1969 method.

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5 8 November 1969: Z340

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Mailed to the Chronicle with a 'dripping pen' greeting card. Many published 'solutions' over the years, none verifiable.

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6 Z13 and Z32

Just show them and move on. Part 1 opens by coming back to Z13: many consistent solutions, and whether the ciphertext can ever pick one.

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7 Plan

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No note.

PART I

Is there a unique answer?

at ≈ 7:00

9 How many answers fit?

2 clicks

Opening question for Part 1. Consistency with the ciphertext is cheap when the ciphertext is short: these 18 all satisfy every constraint Z13 imposes (I checked each against the repeat pattern), and there are thousands more. A codebreaker who finds one consistent plaintext has not proved anything. So before asking how to search for a key, ask whether the ciphertext pins down a unique answer at all, and how long it has to be before it does. Shannon answers that with the unicity distance. Top row is the real Z13, cut glyph by glyph; colors mark repeated symbols and the letters they must share: red S at 1 and 12, blue E at 3 and 11, green E at 5, 7, 9, amber T at 8 and 13.

10 Entropy

$0 \log 0 = 0$. Entropy measures randomness in bits: the average number of yes/no questions needed to pin down X.

11 Markov chains, and language as one

1 click

Time-homogeneous chains only. Aperiodicity rules out e.g. a chain bouncing between two states. The convergence theorem gives long-run letter frequencies (π for English bigrams, which are irreducible and aperiodic), and later says where a sampler ends up. English is not exactly Markov, but this captures the structure that matters. The entropy rate is the average uncertainty in the next symbol given the current one; for i.i.d. symbols it is just $H(X)$.

12 Cipher systems and redundancy

1 click

Shannon, *A Mathematical Theory of Communication* (1948); *Communication Theory of Secrecy Systems* (1949). Before searching, check the problem is well posed: a unique answer exists when the ciphertext carries more information than the cipher system can hide. Shannon developed exactly this theory in the 1949 secrecy paper. A message m and an independent key k are chosen; the codebreaker sees only $c = E_k(m)$. Redundancy measures how far English is from random text.

13 The AEP and typical sequences

1 click

Proof of the AEP: by the Markov property $\Pr(X_0, \ldots, X_n) = \Pr(X_0) \prod_k P_{X_{k-1}X_k}$. Take $-\frac{1}{n}\log$: the $\Pr(X_0)$ term goes to 0, the rest is a time average of $f(i,j) = -\log P_{ij}$. The pair chain (X_{k-1}, X_k) is ergodic with stationary law $\pi_i P_{ij}$, so by the ergodic theorem the time average converges to $\sum_{i,j} \pi_i P_{ij} f(i,j) = H$. Typical set: (1) a.s. convergence implies convergence in probability, so $\Pr(A_\varepsilon^n) \to 1$. (2) Each typical x has $\Pr(x) > 2^{-n(H+\varepsilon)}$ and probabilities sum to at most 1, so $|A_\varepsilon^n| \le 2^{n(H+\varepsilon)}$. (3) For n large $\Pr(A_\varepsilon^n) > 1-\varepsilon$ and each $\Pr(x) < 2^{-n(H-\varepsilon)}$, so $|A_\varepsilon^n| \ge (1-\varepsilon)2^{n(H-\varepsilon)}$. Upshot: a long text is almost surely one of about 2^{nH} roughly equally likely strings, an exponentially small fraction of everything.

14 Unicity distance

2 clicks

Heuristic: wrong decryptions behave like uniform random strings, and a uniform string is typical with probability $|A_\varepsilon^n| \,/\, |S|^n \approx 2^{-n(\log|S| - H)} = 2^{-nD}$.

15 The numbers

In the notes I bounded $54 \log 26 \le 54 \cdot 5 = 270$, giving $U \le 90$; either way 408 is several times larger. We are overcounting keys (every map from 54 symbols to 26 letters) and using English redundancy, which is not exactly Zodiac's. Z13 needs only 13 letters and has fewer than U's worth even for a simple substitution.

PART 2

How do we search for the key? at ≈ 20:00

17 The Metropolis algorithm

No known structure to exploit, so brute force is out. The chain concentrates on keys whose decodings look like English. Reference: Diaconis, The Markov chain Monte Carlo revolution, which opens with exactly this prison-cipher example.

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18 A plausibility model

Smoothing avoids undefined ratios; logs avoid underflow. The better the language model, the faster we expect convergence.

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19 An analogous problem: shuffling by transpositions 1 click

Cutoff, only if asked: for $c < 0$ the walk is still far from uniform, so the transition at $\frac{1}{2} n \log n$ is sharp. a is a universal constant. Metropolis does not accept every transposition, so this is not our chain: it is the analogous chain with a flat target. The proof is Fourier analysis on S_n: Q is a class function so its Fourier transform is scalar on each irreducible, and the $(n-1,1)$ representation dominates.

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20 Why $\frac{1}{2} n \log n$?

Details. Upper bound lemma (Cauchy–Schwarz + Plancherel):

$$\left\| Q^{*k} - U \right\|^2 \le \sum_{\rho \neq \text{ triv}} d_\rho \operatorname{Tr}\left(\hat{Q}(\rho)^k \left(\hat{Q}(\rho)^* \right)^k \right) = \sum_{\lambda \neq (n)} d_\lambda^2 r_\lambda^{2k}$$

Frobenius: $r_\lambda = \frac{1}{n} + \frac{2}{n^2} \sum_{(i,j) \in \lambda} (j - i)$ (box contents). Level-j partitions $(n-j, \mu)$ contribute about $e^{-4cj}/j!$, so the $(n-1,1)$ term dominates and the total is $O(e^{-4c})$, giving $a e^{-2c}$. Lower bound: $\operatorname{fix}(\sigma) = \chi_{(n-1,1)}(\sigma) + 1$ has mean 1 and variance 1 under U; after k steps its mean is $1 + (n-1) r^k \approx 1 + e^{-2c}$, far off when $c \ll 0$. Cutoff intuition, only if asked (coupon collector): each step touches 2 cards; until every card is touched (about $\frac{1}{2} n \log n$ steps), the untouched ones are still fixed points, too many for a uniform permutation.

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21 But we don't run that walk DROPPABLE

First row: Metropolis converges to π, not U. Third row: why the proof breaks. Also, Metropolis refuses some transpositions, so k steps is not k moves.

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22 Where does π peak?

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a, b range over plaintext letters. Strict maximum needs M rich enough to tell f^* apart from every other key. Tempering: same proposals, the flat target deformed into one peaked at f^*. Simulated annealing (used on Z340) varies β over time.

23 A peaked target we can analyze

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Why it is still solvable: the Cayley metric is bi-invariant, so the kernel commutes with conjugation and reduces to conjugacy classes, i.e. partitions. Eigenvalues $\beta_\lambda = (1-\theta) + (\theta n(\lambda^t) + n(\lambda))/\binom{n}{2}$, with $n(\lambda) = \sum_i (i-1)\lambda_i$; eigenvectors come from Jack symmetric functions (Schur functions at $\theta = 1$). Cayley distance d = fewest transpositions from σ to σ_0; Metropolis accepts a move closer to σ_0, otherwise flips a θ-coin. Source: Diaconis, *MCMC revolution*, §3, Theorem 2; the survey prints $f \to 0$ for $c \to 0$, a typo for $c \to \infty$. Caveat: Pl is not a Mallows model and can have local maxima, so for decoding $n \log n$ stays a heuristic, but now one backed by a theorem for a peaked target. Restarts and annealing handle local maxima. In practice a few thousand steps decode English, matching the $n^2 \log n$ scale from a bad start.

PART 3

What made Z340 so much harder?

at ≈ 33:00

25 So it should have been solvable…

2 clicks

Unicity says the answer is determined; search found nothing. That contradiction is itself evidence the model of the cipher is wrong. Cruder bound from the notes: $63 \cdot 5 = 315$ bits, $U \leq 105$; same conclusion. Homophones flatten the histogram: 63 symbols over 340 positions is about 5.4 occurrences per symbol (Z408: about 7.6), so plain frequency analysis sees a nearly flat histogram. Test ciphers: Berg-Kirkpatrick et al. (2013) argued Z340 is not strictly homophonic because Z340-like test ciphers are solved automatically.

26 Errors, or a hidden step?

1 click

No note.

27 Transpositions blow up the key space

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This is the payoff of Part 1: unicity tells you which hypotheses about the cipher are even worth testing. The search over transpositions has to be over a small, guessable family, and the spacing-19 spike says which kind. The 22 bits: von zur Gathen counts $\log_2(26 \cdot 167) \approx 14.7$ bits for splitting into up to 3 sections and $\log_2 208 \approx 7.7$ for transpositions up to 51 places plus word reversals.

28 The clue: period 19

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Fig. 24 and Fig. 26 of the solvers' paper. Nothing in a one-for-one substitution singles out 19, and errors would not create such a spike. Spotted by Van Eycke and 'daikon', independently, summer 2015; both first suspected columnar transposition, which failed. Z340 at period 1 is only 1.36 standard deviations above the shuffle mean; Z408 at period 1 is 7.52 standard deviations above. Test ciphers: 1,000 artificial homophonic ciphers of length 340 with known English plaintexts.

29 The scheme

Full transposition: Fig. 43 of the solvers' paper (`images/transposition.png`). Sections 1 and 2 are $(1, 2)$-decimations. In the last two rows some chunks are reversed. Since $\gcd(9, 17) = 1$ the diagonal walk visits every cell of a 9×17 block exactly once. Procedure: start at position 1, write the symbol, move down one (wrapping), right two (wrapping), repeat.

30 Encoding slips

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Fig. 42. Found by Van Eycke. The reading direction runs down-right through the grey line, so one skipped symbol garbles one letter in each of several words.

31 The search, in practice

Outer loop: a structured, finite family of transpositions. Inner loop: MCMC over homophonic keys. Unicity says the true transposition, once tried, gives a uniquely determined, high-plausibility plaintext.

32 The plaintext

Shown with the original typos corrected; the raw decryption reads e.g. FAN, BRINGO, BECAASE, PARADLCE, SOOHER, WORV (von zur Gathen, p. 3). It continues: WHERE EVERYONE ELSE HAS NOTHING WHEN THEY REACH PARADICE SO THEY ARE AFRAID OF DEATH I AM NOT AFRAID BECAUSE I KNOW THAT MY NEW LIFE IS LIFE WILL BE AN EASY ONE IN PARADICE DEATH.

33 The solution is certified

2 clicks

Closes the loop with Part 1: same argument as for Z408, now for the structured transposition family. Compare $U \approx 890$ for an arbitrary transposition. His formula differs slightly from our $\log K/D$: he treats ciphertext symbols as uniform ($\log_2 63 \approx 5.98$ bits each) and the plaintext as 430 characters including blanks, so $U = I(\text{key})/\left(\log_2 63 - \frac{430}{340} \cdot H\right) = 562.6/\left(5.98 - 1.8 \cdot \frac{430}{340}\right) \approx 152$, with $H = 1.8$ measured on the Zodiac's own letters. Robustness: $H = 1.36$ gives 132, $H = 2.30$ gives 183. Error slack: 25 replacements, $\log_2\left(\binom{340}{25} \cdot 27^{25}\right) \approx 244$ bits.

34 Takeaways

at ≈ 47:00 | click

Peroration: never drop this slide. End here, then stop; the chair takes questions. Open question 1: Diaconis–Hanlon covers only the Mallows model, and Pl is not Mallows and can have local maxima, which is why $n \log n$ stays a heuristic for decoding. Open question 2 is von zur Gathen's closing question; a competing solution would need its own system with unicity distance below 340 (ideally near 152).

35 References

No note.